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Article in *Journal of Innovation Information Technology and Application (JINITA)* · July 2025

DOI: 10.35970/jinita.v7i1.2658

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Hybrid Approach for Protein Secondary Structure Prediction with KNN, SVM, and Neural Network Algorithms

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ARTICLE INFO

Article history:

Received 16 February 2025

Revised 23 May 2025

Accepted 25 June 2025

Available online 30 June 2025

Keywords:

K-Nearest Neighbors

Support Vector Machine

Neural Network

Protein Secondary Structure

Prediction

Hybrid approach.

IEEE style in citing this article:

B. M. Ntumba, J. P. N. Koto-Te-Nyiwa, B. B. Kapesa, and N. K. Mulenda, "Hybrid Approach for Protein Secondary Structure Prediction with KNN, SVM, and Neural Network Algorithms," *Journal of Innovation Information Technology and Application (JINITA)*, vol. 7, no. 1, pp. 68–80, Jun. 2025.

ABSTRACT

One of the main challenges in bioinformatics is predicting the structures of macromolecules, particularly nucleic acids and proteins. In this study, we propose a hybrid approach integrating K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Neural Network (NN) algorithms. We perform an in-depth analysis using various metrics, including accuracy, Q3 score, ROC, and precision-recall curves. Based on the RS126 dataset, we compared our hybrid model with individual approaches, revealing that our model achieves an accuracy of 80% and a Q3 score of 86%, outperforming each of the algorithms separately. These results validate the effectiveness of combining models for protein secondary structure prediction (PSSP). We show that the hybrid model outperforms the other models for this task. We also discuss the implications of these results and propose future work to further improve the accuracy and robustness of the model. This approach could have important implications for protein structure modeling, in particular for understanding their three-dimensional structure and function.

1. INTRODUCTION

Protein secondary structure prediction (PSSP) is an important field in bioinformatics and structural biology, as there is a very close link between the protein's three-dimensional structure and its biological function. Proteins are involved in many biological processes and are essential macromolecules for life. Understanding their secondary structure, mainly composed of alpha-helices (α -helices), beta-sheets (β -sheets), and loops, is a fundamental step towards predicting their tertiary structure [1] [2] [3] [4], their function and consequently mastering drug design, protein engineering, and disease mechanisms [5].

Experimental methods such as X-ray crystallography and NMR spectroscopy provide precise information on protein structures. However, they are expensive, time-consuming, and cannot be applied to all proteins [4] [6] [7]. Yet it is crucial to determine the secondary structure of any protein from its primary structure, even without precise information on local interactions between amino acids, such as hydrogen bonds [8]. This is why machine learning and artificial intelligence are playing an increasingly crucial role in PSSP [8].

Historically, statistical methods such as the Chou-Fasman pattern [9] and the Garnier-Osguthorpe-Robson (GOR) method have been used to predict secondary structures. Although they have provided useful preliminary results, their accuracy is limited by their heuristic and statistical nature [10]. The advent of machine learning has overcome some of these limitations by introducing algorithms such as K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Neural Networks (NN).

In this study, we propose a hybrid approach that combines the strengths of KNN, SVM, and NN to overcome the individual limitations of each algorithm and improve PSSP accuracy. KNN is known for its simplicity and effectiveness on small datasets [11] [12], SVM excels in handling high-dimensional data with clear margins between classes [13] [14], NN captures complex nonlinear patterns in sequences [15] [16]. Their combination ensures robustness, better generalization, and complementary learning dynamics. By exploiting their respective advantages and minimizing their weaknesses, we aim to improve the accuracy, robustness, and interpretability of predictions. This synergistic approach could make significant contributions to structural biology and protein understanding, particularly in contexts requiring reliable secondary structure predictions for downstream applications such as drug discovery, protein engineering, and functional annotation [17].

2. LITERATURE REVIEW

The Chou-Fasman and Garnier Osguthorpe-Robinson algorithms were the first PSSP methods to use statistical approaches based on amino acid frequencies in known secondary structures, paving the way for more advanced approaches involving machine learning [3] [10] [8] [9] [18].

Thanks to machine learning, significant progress has been made since 1994 with the introduction of artificial neural networks, particularly in obtaining accuracy scores. Neural networks have learned from complex models, thanks to large-scale databases containing sequences of proteins and their structures [3] [19].

Recently, the development of deep learning techniques has brought better accuracy scores in PSSP [20]. Convolutional neural networks (CNN) and recurrent neural networks (RNNs), in particular Long Short-Term Memory (LSTM) [21], have shown superior performance. Models such as Deep Convolutional Neural Fields (DeepCNF) combine CNN with Conditional Random Fields (CRF) to capture both local and global features of protein sequences.

Furthermore, to improve the accuracy score in PSSP, tools such as PSIPRED [19] [22] exploit evolutionary information that is crucial for capturing structural constraints conserved during evolution. This evolutionary information is derived from multiple sequence alignments (MSA).

Currently, DeepMind's AlphaFold has developed a very powerful model that has pushed the boundaries of PSSP to new heights, using advanced deep learning architectures and graph modeling techniques [23] [24]. In particular for three-dimensional structure prediction [25] [26]. However, challenges remain, notably structure prediction for non-homologous proteins and the modeling of complex interactions within and between proteins. In addition, although AlphaFold is highly accurate at predicting three-dimensional structure, it is crucial to note that PSSP is faster and more comprehensive at predicting three-dimensional structure. In this context, it can be used in preliminary studies or to analyze large quantities of data [22]. Knowledge of secondary structures can help interpret the results of three-dimensional structure prediction, distinguishing well-predicted regions from less reliable ones [27]. Ultimately, PSSP can also offer higher accuracy in interpreting results for specific protein regions, and can complement the information provided by three-dimensional structure prediction methods [28] [29].

In view of the above, it is worth noting that PSSP has evolved significantly. The technique began with simple statistical methods and has worked its way up to sophisticated deep learning algorithms [25] [30]. These advances have improved the accuracy and reliability of predictions, opening up new avenues for biotechnological research and applications. Future challenges include improving prediction for complex proteins and integrating experimental data to refine existing models.

The interest in proposing a hybrid model is due to the concern for robustness and resilience because the algorithms are combined, and assembly models often outperform individual models. The hybrid model can reduce individual biases and errors by increasing the overall accuracy of predictions. It's also worth noting that hybrid models avoid overfitting on smaller or more varied datasets, and in the same vein, they can adapt and become flexible in the face of varied data such as sequences and structures.

3. MATERIALS AND METHODS

3.1. Dataset

Among the existing datasets, we conducted this study with the RS126 dataset for PSSP. RS126 was first used by Rost and Sander in 1993 and has since been chosen by researchers as a reference for PSSP in bioinformatics [29]. It is composed of annotated and mainly non-homologous protein sequences (not two proteins in this set that share more than 25% sequence identity over a length of more than 80 residues) with their actual secondary structures, which enabled us to draw up an analysis of the performance of the hybrid approach in terms of accuracy, sensitivity, specificity and other metrics [7][31][32]. The proposed model is designed in Python 3.12.4. under Windows and a PC INTEL(R) Core(TM) i3-2350M CPU @2.30GHZ.

3.2. Methods

In this study, we adopted the DSSP (Define Secondary Structure of proteins) representation because it is widely used and allows the secondary structures of eight classes to be converted into three classes as described in Table 1 [33][34][35][36]. With the dataset, we employed several techniques to enable understanding by machine learning models. Firstly, we applied the sliding window technique.

For this purpose, the window is fixed and moves along the protein sequence. At each position, it extracts features that predict the structure of the amino acid at the center. One-hot encoding converts each amino acid into a sparse binary vector (filled mainly with zeros) where a single bit is set to represent its specific presence in the sequence. In the same vein, we have used the principle of orthogonal coding, which creates binary vectors that are mutually orthogonal and provide a distinct and independent representation of each amino acid, as shown in Table 2.

Table 1. DSSP method for reducing states from 8 to 3, to distinguish dictionary elements from protein structures

DSSP Class	8-State Symbols	3-Class Symbols	Element Type	Favorable Amino Acids
Helix α	H	H	Helices	Met, Ala, Leu, Glu, Lys
Helix 3_{10}	G	H		
Helix π	I	H		
Strand β	E	E	Sheets	Thr, Ile, Val, Phe, Tyr, Trp
Bridge β	B	E		
Tower	T	C	Loops	Gly, Ser, Pro, Asp
Bend	S	C		

We also ran the cross-validation test five times, selecting one subset for testing and using the remaining four subsets in training to ensure that each subset was used for both training and validation at different times.

In this study, we have chosen a window of size 17. That is, the sliding window comprises 17 input amino acids, each designated by the value 1 according to its location in the same window as explained above, and all other elements have the value 0. Thus, our input model will have a total of 20 X 17 entries, each entry corresponding to a vector of 20 elements (or bits) to represent the amino acids.

A concrete example is if we have a protein sequence as shown in Figure 1:

AETVESCLAKSHTEN. With a sliding window size estimated at 7, to predict the secondary structure of residue 'S', the window pattern will take the shape TVESCLA so that 'S' ends up in the center to make the prediction.

Then, to progress and predict the structure of residue 'K', the window must slide, leaving out residue 'T', which was at the beginning of its pattern, and adding residue 'K', which is next and closest as the window slides. The pattern thus becomes VESCLAK to predict the central residue 'C'. This process is repeated to run through the sequence step by step, until predictions for all residues are obtained.

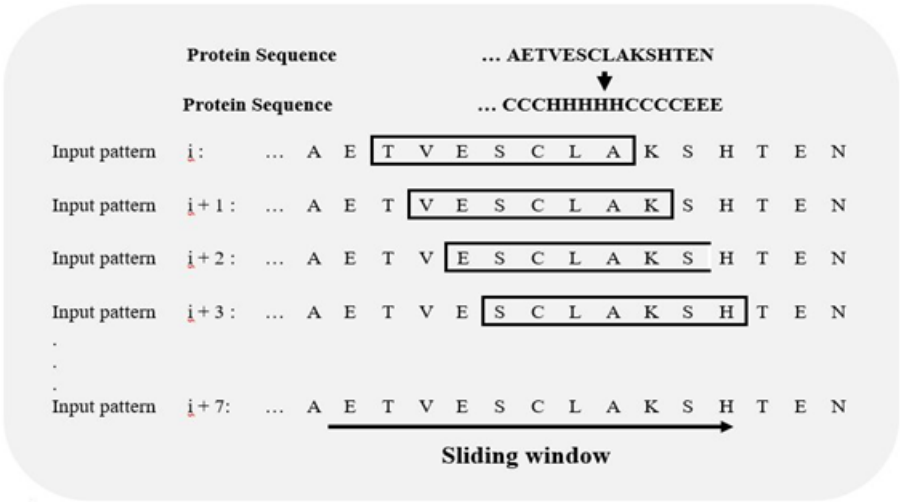


Figure 1. How the sliding window works

Table 2. Orthogonal coding for 15 amino acids

A	E	T	V	E	S	C	L	A	K	S	H	T	E	N
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	1	0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

4. ALGORITHMS USED

Figure 2 shows the architecture of the proposed model in 4 phases. The first phase is called pre-processing data. It covers data cleansing, data normalization or standardization and encoding. The second phase, data vision and algorithm selection, divides the dataset into training and test sets, and enables prediction algorithms to be selected. The third phase, the cross validation and training, allows us to perform cross validation by optimizing hyperparameters and training the model. Finally, the fourth and last phase, called prediction and evaluation, helps us make predictions on the test set, as well as visualizing the results and comparing the performance of each algorithm in the overall model.

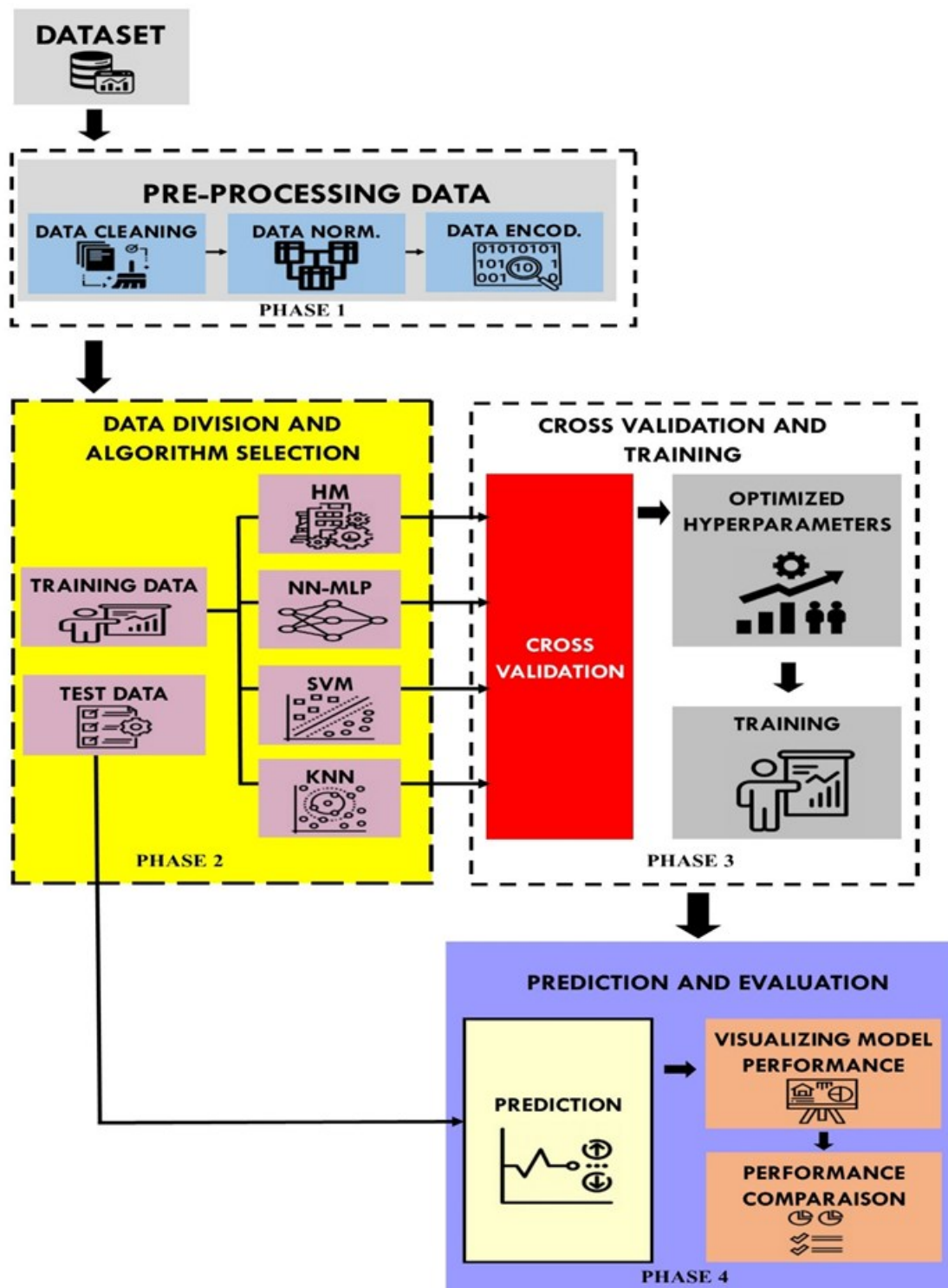


Figure 2. Architecture of the proposed model
HM: Hybrid Model, NN-MLP: Neural Network-Multi Layer Perceptron,
SVM: Support Vector Machine et KNN: K-Nearest Neighbors

4.1. Support Vector Machine (SVM)

SVM is a supervised machine learning algorithm used mainly for classification and regression. Proposed by Cortes and Vapnik in 1995 [14], it seeks to find the hyperplane that best separates classes in a feature space. The hyperplane is chosen to maximize the margin between the nearest data points of two classes (called support vectors). Its mathematical formulation is defined by a data set

$$D = \{(x_i, y_i) \mid i = 1, 2, \dots, n\} \quad (1)$$

where $x_i \in R^d$ represents feature vectors and $y_i \in \{-1, +1\}$ representing labels.

The objective of the SVM is to find the hyperplane defined by:

$$f(x) = w^T x + b \quad (2)$$

that separates classes with maximum margin. SVMs are powerful classifiers that seek to maximize the margin between classes in a high-dimensional feature space. Using specific kernels, SVMs can capture non-linear relationships between amino acids and their secondary structures. Although SVMs are robust and powerful, they require careful hyperparameter tuning.

To enable specialization and better differentiation of secondary structures, we have created six classifiers, each of which will handle the prediction of a specific type of secondary structure. These classifiers are : H/~H: classifies helices versus non-helices, E/~E: classifies sheets versus non-leaflets, C/~C: classifies loops versus non-loops, H/E: differentiates helices from sheets, E/C: differentiates sheets from loops and C/H: differentiates loops from helices. We then chose the kernel function. In the context of our study, we used the RBF (Radial Basis Function) kernel for its ability to handle non-linearity in the data [37]. In the same vein, we focused on the gamma parameter, which we kept constant throughout the experiment. The regularization parameter C, which controls the trade-off between margin and minimization of classification error, was tested with 0.2, 0.4, 0.9, 1 and 4 by optimizing the hyperparameters through grid search and cross-validation.

4.2. Neural Networks (NN)

Neural networks, in particular deep neural networks (DNN) and convolutional neural networks (CNN), have revolutionized protein secondary structure prediction. NNs can learn complex, abstract representations of amino acid sequences, capturing both local and global patterns. Their ability to process large amounts of data and generalize to new sequences makes them ideal candidates for this task. However, NN require significant computational resources and are often regarded as black boxes due to their complexity.

In this work, we used a Feed-Forward Multi-Layer Perceptron artificial neural network. A sliding window of size 17 was applied to capture the local context around each residue. Each amino acid is represented by a one-hot vector of 20 dimensions, giving a combined input of dimensions for each position. This led us to design a network comprising 340 neurons in the input layer, a hidden layer containing 30 neurons, and 3 neurons in the output layer corresponding to the three classes of secondary structures: α -helix (H), β -sheet (E), and loop (C).

4.3. KNN algorithm

In its basic form, the K-nearest neighbors algorithm enables both supervised and nonparametric classification [38]. In this algorithm, a prediction is made between individuals with the same characteristics by defining a distance function that characterizes the proximity between these individuals [39].

Let $x \in R^d$ et $D = \{(x_i, y_i) \mid i = 1, 2, \dots, n\}$ be a training set where $x_i \in R^d$ represents feature vectors and $y_i \in Y$. The objective is to predict $\hat{y}$ for x based on its K nearest neighbors in feature space. In order to measure proximity, we use a distance function d such as the Euclidean distance, for example, which is given by:

$$d: R^d \times R^d \rightarrow R^+$$

$$d(x, x_i) = \sqrt{\sum_{j=1}^d (x_j - x_{i,j})^2} \quad (3)$$

The K training points closest to x are identified by minimizing $d(x, x_i)$. The set of corresponding indices is noted:

$$\mathfrak{K}_k(x) = \operatorname{argmin}_{S \subseteq \{1, \dots, n\}, |S|=K} \sum_{i \in S} d(x, x_i) \quad (4)$$

Thus, for classification, the prediction is :

$$\hat{y} = \operatorname{mode}\{y_i \mid i \in \mathfrak{K}_K(x)\} \quad (5)$$

In bioinformatics, KNN has been used to predict secondary structures by analyzing local similarities between amino acid sequences [10] [40]. However, KNN can be sensitive to noise and data dimensionality, which limits its performance [38]. Table 3 shows the KNN algorithm we used for the PSSP problem. We considered the protein sequence to be predicted, then defined a list of training sequence fragments, the size of the sliding window ($n = 17$), and the number of neighbors ($k = 50$).

4.4. Hybrid approach

The proposed architecture combines KNN, SVM, and NN via a majority voting classifier. Each algorithm is trained separately, and then its predictions are merged. In the event of a tie, a weighting strategy based on the previous performance of each model is considered to avoid an arbitrary choice [41] [42]. The architecture of the proposed model is divided into four distinct phases, as illustrated in Figure 2.

a. Structure of the hybrid approach

- Phase 1: Pre-processing data

This phase encompasses data cleansing, normalization or standardization, and encoding. For amino acid encoding, the “one-hot encoding” technique is applied, converting each amino acid into a hollow binary vector. The principle of orthogonal encoding is also used to create mutually orthogonal binary vectors, ensuring a distinct and independent representation of each amino acid, as detailed in Table 2. A sliding window technique is employed, where a fixed-size window moves along the protein sequence, extracting features to predict the structure of the central amino acid. For this study, a window of size 17 was chosen, meaning that 17 input amino acids are taken into account, each represented by a value of 1 at its location and 0 elsewhere, resulting in a total of 20×17 entries.

- Phase 2: Data division and algorithm selection

This stage consists of dividing the dataset into a training set and a test set, then selecting the prediction algorithms: KNN, SVM, Neural Networks (NN-MLP), and Hybrid Model (HM).

- Phase 3: Cross-validation and training

This phase is dedicated to cross-validation, hyperparameter optimization, and model training. To assess the robustness and generalizability of the model, a 5-fold cross-validation test was carried out. This technique ensures that each subset of the dataset is used for both training and validation at different times, reducing the risk of overlearning and providing a more reliable estimate of model performance.

- Phase 4: Prediction and evaluation

The last phase is devoted to prediction on the test set, visualizing the results, and comparing the performance of each algorithm making up the hybrid model. Following training and optimization, prediction is performed on an independent test set to evaluate the performance of the algorithms and the hybrid model.

b. Model combination mechanism

The hybrid approach combines the predictions of the KNN, SVM, and NN algorithms via a majority vote classifier. Each algorithm is trained independently, and its predictions are then merged. In the event of a tie, a weighting strategy based on the previous performance of each model is implemented to avoid an arbitrary choice.

- Majority voting formula

Majority voting is formulated as follows: Let M be the number of basic models, where each model $m_i (i = 1, 2, \dots, M)$ generates a prediction $\hat{y}$ for a dataset x and C the set of possible classes. With $\hat{y}_i$ The prediction of the model m_i for data x .

$$votes_c = \sum_{i=1}^M I(\hat{y}_i = c) \quad (6)$$

Where $(\hat{y}_i = c)$ is a function that indicates 1 if the model prediction m_i is equal to class c , otherwise it is 0.

- Final prediction

The class c^* with the highest number of votes is chosen as the final prediction of the majority voting model:

$$\hat{y}_{final} = votes_c \quad (7)$$

For example, suppose we have three models whose predictions are: $\hat{y}_1 = 1$, $\hat{y}_2 = 0$ et $\hat{y}_3 = 1$. The votes for class 0 would be 1, and for class 1 would be 2, leading to a final prediction of 1. In the event of a tie, the model uses a weighting mechanism.

$$votes_0 = \sum_{i=1}^3 I(\hat{y}_i = 0) = 1 \quad (8)$$

$$votes_1 = \sum_{i=1}^3 I(\hat{y}_i = 1) = 2 \quad (9)$$

The class $c^* = 1$ is the final prediction of the majority vote. So $\hat{y}_{final} = 1$. In the event of a tie, our model uses a weighting mechanism to determine the appropriate class.

5. RESULTS

Table 4 shows the performance (Accuracy and Q3) of the individual algorithms (KNN, SVM, NN-MLP) and the hybrid model (HM), enabling a direct comparison of their efficiency.

Table 4. Metrics

ALGORITHMS	ACCURACY	Q ₃
KNN	55%	63%
SVM	71%	78%
NN-MLP	63%	60%
HM	80%	86%

We observe strong empirical evidence that the proposed hybrid approach performs best for protein secondary structure prediction among the models tested in this study. The integration of the three algorithms (KNN, SVM, NN-MLP) via a majority voting system achieves superior accuracy, validating the effectiveness of this strategy for the PSSP problem. This justifies the value of model fusion for improving the robustness and accuracy of prediction systems in bioinformatics.

The Calibration curve for the hybrid model, shown in Figure 3, measures the correspondence between predicted probabilities and observed frequencies.

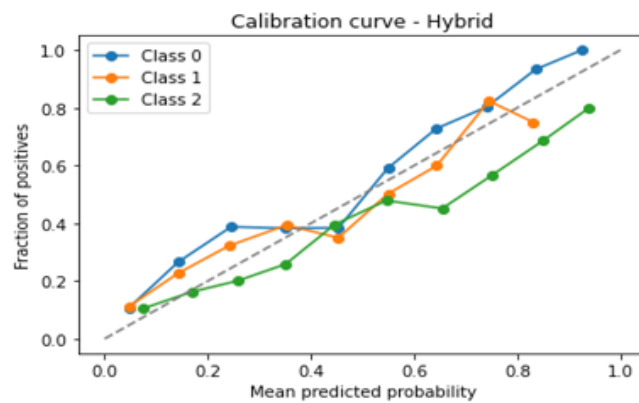


Figure 3. Calibration curve for the hybrid

It indicates that the hybrid model is reasonably better calibrated, like the NN-MLP too, but on the other hand, in Figures 4 and 5, we observe that the SVM and KNN, respectively, present calibration problems, limiting their reliability for probabilistic predictions in our context.

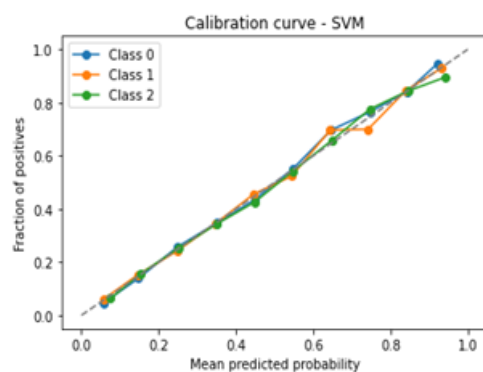


Figure 4. SVM calibration curve

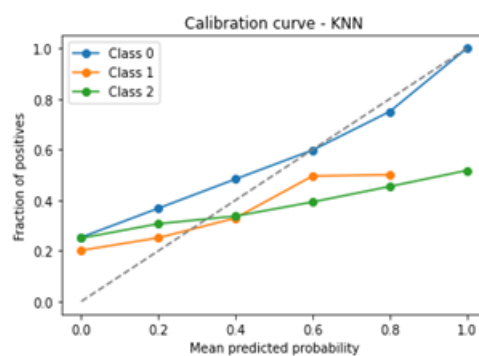


Figure 5. Calibration curve for KNN

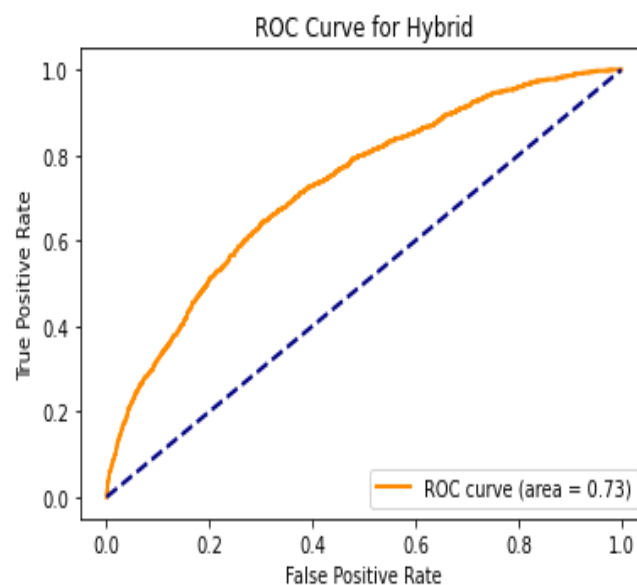


Figure 6. Calibration curve for the hybrid model

ROC curves and areas under the curve (AUC: Area Under the Curve) illustrate the models' overall discriminative ability. Particularly for the ROC curve of the hybrid model shown in Figure 6, the AUC is 73%. This confirms that this model offers the best ability to distinguish between classes.

Similarly, Figures 7 and 8 show the Precision-Recall curves of the NN-MLP and the hybrid model, respectively, in order to measure the correspondence between predicted probabilities and observed frequencies. The hybrid model shows the best overall performance, particularly for class 0. The MLP showed moderate but consistent performance.

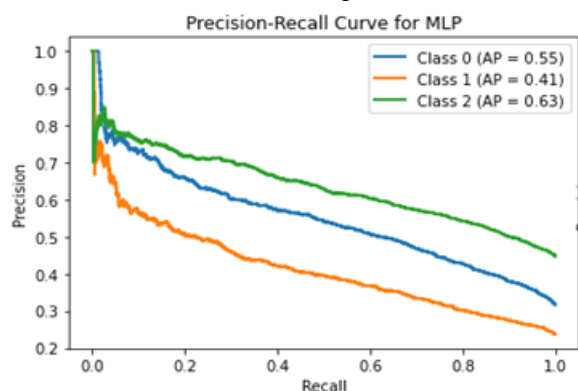


Figure 7. NN-MLP Precision-Recall curve

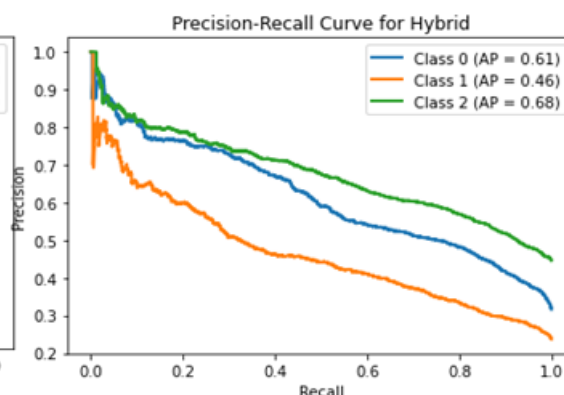


Figure 8. Hybrid model Precision-Recall curve

As shown in Figure 9 for the sequence of human Myoglobin, we observe that each of the models simulates a prediction where the predominance of helices (H) is evident, which is consistent with the known tertiary structure of myoglobin. Loops (C) are also predicted at certain locations, reflecting transition regions between helices or helix-disrupting residues (such as proline, for example). Note that sheet prediction (E) is very rare, which is expected for myoglobin, which is an almost exclusively alpha-helical protein.

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=====
Protein Secondary Structure Prediction Program (Hybrid Model)
=====

This program illustrates the robust prediction of protein secondary structure
by integrating the 'decisions' from three distinct algorithmic approaches:
K-nearest Neighbors (KNN), Support Vector Machine (SVM), and Neural Networks (NN).

Primary Sequence for Analysis : VLSEGEHLVLMVAKVEADVAGHGQDILIRLFKSHPETLEKFDKRF
Processing sequence and generating predictions...

=====
Predicted Secondary Structure Summary
=====
Primary Sequence      : VLSEGEHLVLMVAKVEADVAGHGQDILIRLFKSHPETLEKFDKRF
KNN Prediction        : E H C H C H E H H E H H E H H E H H C E H C H C E H H E H H E H C E
SVM Prediction        : H H C H C H H H H H H H H H H H H H C H H H H H H H H H H H H H H
NN Prediction         : C H C H C H C H C H C H H H C H C C C C C C C H H C C H H C C H H C C
HYBRID Model Prediction : H H C H C H H H H H H H H H H H H C H C H C H H H H H H H C H H H H H C H
=====

Legend for Secondary Structures: H = Helix, E = Strand, C = Coil.

```

Figure 9. Prediction capture

The prediction of our hybrid approach shows the result of the majority vote between the three models. We can see that the hybrid prediction tends to consolidate the majority predictions of the individual models. In most cases, if at least two models predict "H", the hybrid model also predicts "H". This fusion approach reduces the variance of predictions and increases confidence in the final result, as discussed in the work of Burka et al. [41] and Zantedeschi et al. [42] on majority voting in machine learning. The hybrid model's prediction thus reflects a consensus, aimed at producing a more robust and reliable secondary structure.

This demonstration highlights the conceptual advantage of the hybrid approach: by combining the strengths of multiple algorithms, the final model is capable of generating more stable and potentially more accurate predictions than any model taken in isolation, even in this simplified illustration. This principle lies at the heart of our proposal to improve protein secondary structure prediction.

So, even in terms of metrics, AUC, precision, and recall, the results still show that the hybrid model and its approach are the most perceptive. This is due to its ability to combine the strengths of the

individual models. However, some challenges remain, particularly in terms of model performance, which varies between classes. This could reflect an imbalance in the data.

6. DISCUSSIONS

This study demonstrated that the hybrid model combining KNN, SVM, and NN algorithms consistently outperformed the individual models in protein secondary structure prediction. With an accuracy of 80% and a Q3 value of 86%, the hybrid model showed a significant improvement over the isolated performance of the other models. This superiority is attributed to the complementary nature of the algorithms used, each contributing differently to the classification of secondary structures (α -helices, β -sheets, and loops).

The performance obtained corroborates the observations of previous work, particularly that using ensemblistic models for secondary structure prediction. For example, approaches based on CNNs and RNNs have shown significant improvements, but their use is often associated with increased computational complexity. Compared with models such as AlphaFold, although highly accurate for three-dimensional prediction, our approach stands out for its speed and ability to operate efficiently on preliminary data or large ensembles, making it more suitable for exploratory studies.

The results of this hybrid model open up new perspectives in bioinformatics, particularly for rapid drug design and protein functional analysis. This ability to improve the accuracy of secondary structure prediction can serve as a basis for more complex studies of protein interactions or three-dimensional modeling as observed in Table 5.

Table 5. Comparison of models

MODÈLE	ACCURACY (%)	Q3 (%)
HM (KNN, SVM, NN)	80	86
PSIPRED (ensemblist model)	~77	~83
DeepCNF (CNN + CRF)	~79	~84
AlphaFold (3D + PSSP)	~88	-
BAT-FLANN	~75	-
RNN (LSTM)	~78	-

Despite overall satisfactory performance, certain limitations remain, notably calibration problems for KNN and SVM, class imbalance, and computational complexity. We note that the calibration curves indicate a lower reliability of these models for probabilistic predictions; in addition, the variations in performance between classes suggest an imbalance in the RS126 data set. And, although reduced compared to models like AlphaFold, the integration of multiple algorithms can increase resource requirements.

7. CONCLUSION

This study highlights the superiority of the hybrid model for multi-class classification. By integrating the predictions of several models, it achieves high performance in terms of metrics, AUC, and calibration. By integrating the strengths of three complementary algorithms, this hybrid model sets a new benchmark for rapid and accurate protein secondary structure prediction. It represents a significant advance for bioinformatics applications, while also pointing the way to promising areas of improvement and development.

8. FUTURE RESEARCH

For future research, we'll be focusing on improving the data by incorporating techniques such as oversampling or class balancing to reduce biases linked to data imbalance. We'll also be looking at optimizing hyperparameters. That is, exploring more advanced strategies such as genetic algorithms or Bayesian optimization. In addition, advanced hybrid models incorporating deep learning approaches

maximize performance. By validating them on diverse sets, we can combine these techniques to achieve better results.

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